Relational Model Concepts and Integrity Constraints

In relational databases, data is organized into Relations (tables), Tuples (rows), and Attributes (columns). Each attribute has a defined Domain of valid atomic values.

Relational Integrity Constraints guarantee data validity:
1. Entity Integrity: Primary key values cannot be NULL, ensuring each tuple is uniquely identifiable.
2. Referential Integrity: A foreign key in a referencing relation must either match a valid primary key in the referenced relation or be completely NULL.
3. Domain Integrity: Attribute values must conform to defined data types, formats, and check constraints.

Keys Classification and Relational Algebra

Keys enforce relational uniqueness and relationship integrity:

1. Super Key: Any set of attributes that uniquely identifies a tuple within a relation.
2. Candidate Key: A minimal super key (no proper subset is a super key).
3. Primary Key: The chosen candidate key used by the database system as the primary tuple identifier.
4. Alternate Key: Candidate keys not chosen as the primary key.
5. Foreign Key: An attribute referencing the primary key of another relation.

SQL Operations and Join Execution Mechanics

SQL categorizes operations into Data Definition Language (DDL: CREATE, ALTER, DROP) and Data Manipulation Language (DML: SELECT, INSERT, UPDATE, DELETE).

Join Operations combine rows from two relations based on predicate evaluation:
- Inner Join: Returns rows with matching keys in both relations.
- Left Outer Join: Returns all rows from the left relation, padding unmatched right attributes with NULL.
- Right / Full Outer Join: Preserves unmatched rows from right or both relations.
- Cross Join: Computes Cartesian product (size |R| * |S|).